STATISTICAL INFORMATION THEORY — CHEATSHEET (side 1: Lectures 1–4)

LECTURE 1 · INTRODUCTION & PROBABILITY RECAP

- **Info theory = art of redundancy**: *remove* it when it hurts (compression/source coding), *add* it when it helps (channel coding). Info & statistics = two sides of one coin. Course: source coding → channel coding → statistics/geometry.
- **Source coding**: map data symbols → {0,1}; want *short* + *easily decodable*; frequent symbols get shorter words (e.g. B→0, R→10, W→11: 1.56 bit/px vs 3 one-hot).
- **Channel**: **BSC** flips bit w.p. *f*. Repetition code R_3: 0→000, 1→111, r = t + n; optimal decoder = Bayes: $\hat{s} = \text{argmax}_s\, P(r|s)P(s)$ (here majority vote). $p_b = 3f^2(1-f) + f^3 \approx 3f^2$: error O(f)→O($f^2$) but rate ÷3. Shannon: error-free comms at **non-zero rate** is possible (rate < C).
- **Logs**: log=$\log_2$, ln=$\log_e$; log(xy)=log x+log y; log(x^n)=n log x; $\log_b x = \log_a x / \log_a b$; log 1=0; log(x+y) does NOT simplify; ln2≈0.69; log(1/8)=−3.
- **Convex** ("convex-down", ∪): $f(\lambda x_1+(1-\lambda)x_2) \le \lambda f(x_1)+(1-\lambda)f(x_2)$; prove via $f''\ge 0$ or convexity-preserving ops (affine map f(Ax+b); non-neg. weighted sum). log and −x log x are **concave** (∩).
- **RV** X: Ω→E measurable fn. P(ω)≥0, P(Ω)=1. Discrete: pmf, Σp=1. Continuous: pdf, ∫p=1 (*density can be >1*). Notation: X rv, x realisation, $\mathcal{X}$ alphabet.
- **Expectation** $E_p[f(X)] = \Sigma_x\, p(x)f(x)$; Var = $E[X^2]-\mu^2$. **i.i.d.** = independent + identically distributed.
- **Jensen** (f convex ∪): $E[f(x)] \ge f(E[x])$; (concave ∩ reverses: E[log x] ≤ log E[x]).
- **Gaussian** $N(\mu,\Sigma)$: $p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-\tfrac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu))$; 68.3/95.5/99.7% within 1/2/3σ.

LECTURE 2 · ENTROPY & SOURCE CODING THEOREM

$h(x) = \log 1/p(x)$ (surprisal) | $H(X) = \Sigma\, p(x)\log 1/p(x) = E[h(X)]$

- Units: $\log_2$ → bits; ln → nats. Binary entropy $H_2(p) = -p\log p - (1-p)\log(1-p)$ (max 1 bit at p=½). Convention **0 log 0 = 0**. Unlikely outcomes carry more info.
- **Bounds**: $0 \le H(X) \le \log|\mathcal{X}|$. Min: Kronecker δ ($p_i$=1). Max: uniform. Independent X,Y: **h and H are additive**: H(X,Y)=H(X)+H(Y).
- **Shannon axioms** for surprisal i(p): (1) i(1)=0; (2) di/dp ≤ 0 (rarer = more surprising); (3) i(pq)=i(p)+i(q) ⇒ only $i(p) = -\log_b p$.
- Guessing games: uniform ⇒ h(x)=#yes/no questions = binary-encoding length (63-game: 6 bit). Submarine: total info always log 64 = 6 bit however many misses (Σ log(64/63)+…+log(n+1)/n + log n). Info = probability-mass exclusion (Hartley). **Design experiments with evenly probable outcomes.**
- **Differential entropy** $H(X) = -\int f\log f\,dx$. Derived from bins Δ: $H(X^\Delta) = -\Sigma\,\Delta f\log f - \log\Delta$ (diverges as Δ→0; drop it). Can be **negative**; U[0,a]: H = log a (2^H = volume).
- **Gaussian**: h(x) = ln(σ√2π) + ½((x−μ)/σ)² (≈ scaled MSE: lower by moving μ→x or (sometimes) raising σ).

$H(N(\mu,\Sigma)) = \tfrac{1}{2}\ln|2\pi e\Sigma|$ (1D: $\tfrac{1}{2}\ln 2\pi e\sigma^2$)

- Entropy = generalised variance; ≈ log-volume of iso-probability contours.

SOURCE CODING THEOREM (SCT)

- **Code** C: $\mathcal{X}\to\mathcal{D}^*$; ℓ(x) codeword length; L=E[ℓ]. $|\mathcal{D}|$=2 binary, d-ary. Raw bit content = $\log|\mathcal{X}|$ (use $\lceil\log|\mathcal{X}|\rceil$, "extra bit" if not power of 2).
- **WLLN**: $\bar{Y}_n = (1/n)\Sigma X_i \to \mu$ in prob.; E[$\bar{Y}$]=μ, Var($\bar{Y}$)=σ^2/n. High-dim Gaussian: points in a thin shell, Std(r^2)/E[r^2]=O(1/√n)→0.
- **AEP**: $X_1..X_n$ i.i.d. ⇒ $-(1/n)\log p(x_1..x_n) \to H(X)$ (by LLN). All typical samples ≈ equiprobable.
- **Typical set** $A_\varepsilon^{(n)}$: $2^{-n(H+\varepsilon)} \le p(x^n) \le 2^{-n(H-\varepsilon)}$. **(1)** $x^n\in A \Rightarrow |-(1/n)\log p(x^n) - H| \le \varepsilon$, $p\approx 2^{-nH}$; **(2)** $P(x^n\in A) > 1-\varepsilon$; **(3)** $|A| \le 2^{n(H+\varepsilon)}$; **(4)** $|A| \ge (1-\varepsilon)2^{n(H-\varepsilon)}$. So $|A|\approx 2^{nH}$ out of $|\mathcal{X}|^n$. (Proofs: 3 by $\Sigma p=1\ge|A|\cdot 2^{-n(H+\varepsilon)}$; 4 by $1-\varepsilon<P(A)\le|A|2^{-n(H-\varepsilon)}$.) Biased coin p: typical strings have ≈ fraction p of 1s (binomial vs $p^r(1-p)^{n-r}$ tradeoff). Entropy = fraction of log-volume occupied: $\log|A|/\log|\mathcal{X}|^n = H/\log|\mathcal{X}|$.

SCT: for any ε>0, large n ⇒ ∃ invertible code with $E[\ell(X^n)/n] \le H(X)+\varepsilon$. Converse: no invertible code has $L < H(X)$.

(achievability + converse.)

- **Achievability proof**: flag bit 0 + index in A ($\ell \le 1+\lceil\log|A|\rceil \le n(H+\varepsilon)+2$) else flag bit 1 + index in $\mathcal{X}^n$ ($\ell \le n\log|\mathcal{X}|+2$); $L \le n(H+\varepsilon')$, $\varepsilon'=\varepsilon+\varepsilon\log|\mathcal{X}|+2/n$. **Converse**: sub-typical set C with $|C|\approx 2^{n(H-\kappa)}$ has $P \approx 2^{-n\kappa}\to 0$.

LECTURE 3 · SYMBOL CODES, KL DIVERGENCE

- **No free lunch**: no compressor shortens every string. Lossy (maps some strings together) vs lossless (shortens most, lengthens some). Block codes need infinite blocks → use symbol codes.

Code type	Definition
Nonsingular	x≠x′ ⇒ C(x)≠C(x′) (symbol-level invertible)
Extension C*	$C^*(x_1..x_n)=C(x_1)\ldots C(x_n)$
Uniquely decodable	C* nonsingular (Morse: nonsingular, *not* UD)
Prefix (instantaneous)	no codeword is a prefix of another; tree leaves = codewords

Prefix ⊂ UD ⊂ nonsingular. Invertible ≠ easy to invert (RSA: a·b). Adding a word removes all its descendants.

Kraft: prefix code $\Leftrightarrow \Sigma_i D^{-\ell_i} \le 1$ (also holds for all UD codes)

Proof: ℓ_{max} tree has ≤ $D^{\ell max}$ leaves; word of length ℓ_i owns $D^{\ell max-\ell_i}$ of them ⇒ $\Sigma D^{\ell max-\ell_i} \le D^{\ell max}$. "Codeword supermarket": short words cost more; budget =1. Satisfying Kraft means a prefix code with those lengths exists.

OPTIMAL CODES (LAGRANGE)

Lagrange: min f s.t. g=0 ⇒ $\mathcal{L}$=f+λg, $\nabla_{x,\lambda}\mathcal{L}$=0. Min $\Sigma p_i\ell_i$ s.t. $\Sigma D^{-\ell_i}$=1: $\partial\mathcal{L}/\partial\ell_i = p_i - \lambda D^{-\ell_i}\ln D = 0 \Rightarrow \lambda = 1/\ln D \Rightarrow$

$\ell_i^* = -\log_D p_i \Rightarrow L^* = H_D(X)$ (entropy = min avg length of any symbol code)

Needs ℓ integer in practice ⇒ round up $\lceil -\log_D p_i\rceil$ (assumed continuous & equality in derivation). Duality: $p_i \propto D^{-\ell_i}$ = "implicit probabilities" of a code.

- **Huffman**: repeatedly merge the 2 least-probable symbols into one; read codes off the tree. Optimal: L(C*) ≤ L(C′) for any UD code. Ex: p={.25,.25,.2,.15,.15} → ℓ={2,2,2,3,3}, L=2.3.

WRONG CODE · CROSS-ENTROPY · KL

$H(p,q) = -\Sigma\, p\log q$ (len. of q-optimal code on p-data)
$D_{KL}(p\|q) = \Sigma\, p\log(p/q) = H(p,q) - H(p) \ge 0$ (extra cost)

- **Gibbs' inequality**: $D_{KL} \ge 0$, =0 iff p=q. Proof: $-D = \Sigma p\log(q/p) \le \log\Sigma\, p\cdot q/p = \log 1 = 0$ (Jensen, log concave); convex in q ⇒ unique minimum at p=q.
- KL is **asymmetric** (not a distance); =∞ if ∃x: q(x)=0, p(x)>0. D(p‖u) = log|$\mathcal{X}$| − H(p) ⇒ H ≤ log|$\mathcal{X}$|. Ex: $D(p_1\|p_2)$=0.25, $D(p_2\|p_3)$=∞, $D(p_3\|p_2)$=0.5, $D(p_3\|p_1)$=1 bit. Continuous KL ≥0 too (log Δ cancels).
- **Gaussians**: $D_{KL}(p\|q) = \tfrac{1}{2}[\sigma^2_p/\sigma^2_q - 1 - \ln(\sigma^2_p/\sigma^2_q) + (\mu_p-\mu_q)^2/\sigma^2_q]$; $H(p,q) = \tfrac{1}{2}\ln 2\pi\sigma^2_q + [\sigma^2_p+(\mu_p-\mu_q)^2]/2\sigma^2_q$. Narrow Gaussians: KL grows faster with Δμ.
- **MLE**: θ* = argmax p(x;θ); log-lik $\mathcal{L}$=(1/n)Σ ln $p(x_i;\theta) \to E_p[\ln p(x;\theta)]$ (n→∞) ⇒ **ML = min KL = min code length**. Gaussian mean: $\mu^*=(1/n)\Sigma x_i$.
- **Good compressor = good predictor**: code ℓ ⇒ $q=2^{-\ell}/z$, $z=\Sigma 2^{-\ell}$ (Kraft number ≤1). Extra cost $E_p[\ell-\ell^*] = \log(1/z) + D_{KL}(p\|q) \ge 0$; zero iff Kraft equality & p=q.

LECTURE 4 · JOINT ENTROPY & MUTUAL INFORMATION

- **Joint RVs** = different partitions of Ω; p(x,y)=measure of $X^{-1}(x)\cap Y^{-1}(y)$. Marginal $p(x)=\Sigma_y p(x,y)$; conditional p(x|y)=p(x,y)/p(y). Product p(x,y)=p(x|y)p(y); sum $p(x)=\Sigma p(x|y)p(y)$; Bayes p(x|y)=p(y|x)p(x)/p(y).
- **Independence**: $X\perp Y \Leftrightarrow p(x,y)=p(x)p(y)$ (⇒p(x|y)=p(x)); cond.: $X\perp Y|Z \Leftrightarrow p(x,y|z)=p(x|z)p(y|z)$. Conditioning can create/destroy independence: Y=f(X),Z=g(Y): X⊥Z but X⊥Z|Y; Z=X⊕Y (fair coins): X⊥Z but X⊥Z|Y.
- **Bayesian network**: $p(X)=\prod_k p(X_k|pa_k)$. **Markov chain** X–Y–Z: $p(X_t|X_{t-1}..X_1)=p(X_t|X_{t-1})$ (⇒X⊥Z|Y); order-p: depends on last p.

$H(X,Y) = -\Sigma\, p(x,y)\log p(x,y)$ $\quad H(X|Y) = \Sigma_y\, p(y)H(X|Y{=}y)$
Chain rule: $H(X,Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$

- H(X|Y) ≥ 0 (disc.); ≤ log|$\mathcal{X}$|; =0 if X=f(Y); =H(X) if X⊥Y. **Invariance**: discrete H invariant under bijections (H(X|Y)=H(Y|X)=0 ⇒ H(X)=H(Y)).
- **Cond. KL**: $D(p(X|Y)\|q(X|Y)) = \Sigma_y\, p(y)\, D(p(X|y)\|q(X|y)) \ge 0$.
- **Stationary process** ($p(X_1..X_t)=p(X_{1+s}..X_{t+s})$). **Entropy rate** h = lim $(1/n)H(X_1..X_n)$ = lim $H(X_n|X_1..X_{n-1})$ (equal for stationary); conditioning on more history never increases it; reaches h at Markov order p. Coin-in-a-box (flip w.p. p): h = $H_2(p)$. Stream codes (Lempel–Ziv/zip) use entropy rate.

$I(X;Y) = \Sigma\, p(x,y)\log[p(x,y)/(p(x)p(y))] = D_{KL}(p(X,Y)\|p(X)p(Y))$
$= H(X)+H(Y)-H(X,Y) = H(X)-H(X|Y) = H(Y)-H(Y|X)$

- Symmetric; **I ≥ 0**; **I=0 ⇔ X⊥Y**; I(X;Y) ≤ min{H(X),H(Y)} (discrete). **Info doesn't hurt**: H(X|Y) ≤ H(X) (but H(X|Y=y) may exceed H(X)).
- **Cond. MI** I(X;Y|Z)=H(X|Z)−H(X|Y,Z)=D(p(X,Y|Z)‖p(X|Z)p(Y|Z)) ≥ 0; =0 ⇔ X⊥Y|Z. **MI chain rule**: I(X,Y;Z) = I(X;Z) + I(Y;Z|X).
- **Data processing inequality**: X–Y–Z ⇒ I(X;Z) ≤ I(X;Y) (I(X;Y,Z)=I(X;Z)+I(X;Y|Z)=I(X;Y)+I(X;Z|Y), last =0). MI invariant under bijections (even continuous).
- **Venn**: H(X)=μ(A), H(X,Y)=μ(A∪B), H(X|Y)=μ(A\B), I=μ(A∩B). Co-information I(X;Y;Z)=H(XYZ)−H(XY)−H(YZ)−H(XZ)+H(X)+H(Y)+H(Z) can be **negative**.
- **Gaussian MI**:

$I(X;Y) = -\tfrac{1}{2}\log(1-\rho^2) = \tfrac{1}{2}\log(\sigma_{yy}/\sigma^2_{y|x})$

; uncorrelated ≠ independent, MI captures non-linear dependence.
- **MI = bound on prediction**: I(X;Y) = H(Y)+E[log p(y|x)] ≥ $H(Y) + E_{p(x,y)}[\log q_\theta(y|x)]$ (Gibbs); equality if p(y|x)∈Q at best θ. Mock exam 2(c): empirical avg $(1/n)\Sigma \log q(y_i|x_i) \approx E[\log q]$.

STATISTICAL INFORMATION THEORY — CHEATSHEET (side 2: Lectures 5–8)

LECTURE 5 · CHANNEL CODES

- **DMC**: input alphabet $\mathcal{X}$, output $\mathcal{Y}$, p(y|x); extended channel $p(y^n|x^n)=\prod p(y_i|x_i)$. **BSC(f)**: I(X;Y)=H(Y)−H(Y|X)=1−H_2(f) for uniform input (H(Y|X)=H_2(f)).

Capacity: $C = \max_{p(x)} I(X;Y)$ (optimal $p^*(x)$ = capacity-achieving input)
Recipe: $I = H(Y) - H(Y|X)$; $H(Y|X)=\Sigma\, p(x)H(Y|X=x)$; parametrise p(X=1)=α, maximise over α∈[0,1]

- **Noisy typewriter** (27 symbols, each → itself/2 neighbours w.p. ⅓): H(Y)=log27, H(Y|X)=log3, I=log9; use every 3rd letter ⇒ non-confusable inputs (disjoint output sets) ⇒ error-free. Channel AEP: $2^{nH(Y)}$ outputs, $2^{nH(Y|X)}$ per input ⇒ $2^{n(H(Y)-H(Y|X))} = 2^{nI}$ distinguishable messages. **Interpretation of C**: max rate (bits/channel use) for reliable (arbitrarily small error) communication.
- **(n,K) block code**: 2^K codewords of length n; **rate R=K/n**. Decoder $y^n \to \hat{s}$. Mean block error $p_B=\Sigma p(s)p(\hat{s}\neq s|s)$; maximal $p_{BM}=\max_s p(\hat{s}\neq s|s)$; bit error p_b.

Noisy-channel coding thm: (1) ∀ε>0, R<C ∃ code (large n) with $p_{BM}<\varepsilon$. (2) With p_b tolerated, rates up to $R(p_b) = C/(1-H_2(p_b))$ achievable. (3) Rates above $R(p_b)$ NOT achievable.

- **Proof ingredients**: (i) **joint typicality** $J_\varepsilon^{(n)}$: x^n typical, y^n typical, (x^n,y^n) typical (each within ε of H(X), H(Y), H(X,Y)). Props: ① $p(x^n,y^n)\approx 2^{-nH(X,Y)}$; ② P(J)→1 (union bound on complements); ③ $|J| \le 2^{n(H(X,Y)+\varepsilon)}$; ④ independent $\tilde{x}^n,\tilde{y}^n \sim p(x^n)p(y^n)$: $P(\in J) \le \mathbf{2^{-n(I(X;Y)-3\varepsilon)}}$. (ii) **random codes** (sample 2^{nR} words from $p(x^n)$; average error; "weighing babies": avg <δ ⇒ some code <δ). (iii) **typical-set decoder**: success iff exactly one s with $(x^n(s),y^n)$ jointly typical. Errors: true $x^n(s)$ not JT (→0 by ②) or another s′ JT: $\le (2^{nR}-1)2^{-n(I-3\varepsilon)} \le 2^{-n(I-R-3\varepsilon)} \to 0$ iff R<I−3ε (use $p^* \Rightarrow$ R<C−3ε). (iv) **expurgation**: drop worst half of words ⇒ $p_{BM} \le 2p_B$, rate R−1/n.
- **Error detection vs correction**: R_2 detects 1 flip (01→?), R_3 corrects 1 flip (101→1).
- **Parity** = XOR = sum mod 2. **Hamming (7,4)**: $n=2^m-1=7$, K=4 data, m=3 parity; 16 codewords; parity bits at positions 1,2,4 (powers of 2): $p_1 d_1 p_2$... layout $p_1 p_2 d_1 p_3 d_2 d_3 d_4$.
  $p_1=d_1+d_2+d_4$, $p_2=d_1+d_3+d_4$, $p_3=d_2+d_3+d_4$ (mod 2).

Symbol	Meaning
$s \in B^K$	source (data) bits
$t = G^T s$ mod 2	transmitted (n bits), $G\in B^{K\times n}$ generator
r = t + n mod 2	received = transmitted + noise
z = H r mod 2	syndrome, $H\in B^{m\times n}$ parity-check

H = [1010101 / 0110011 / 0001111] (column j = j in binary, LSB first). Ht=0 for any codeword; H(t+n)=Hn ⇒ **syndrome = binary position of the flipped bit** (error in bit 3 ⇒ z=110 ⇒ $1\cdot2^0+1\cdot2^1+0\cdot2^2=3$) ⇒ flip it back. G: data columns = I_K; parity columns = data rows of H^T. G=[1110000/1001100/0101010/1101001]. Systematic = data first, parity last (lecture used non-systematic).

- **Hamming distance** d = # differing bits; Hamming code min distance 3. Min distance t corrects $\lfloor (t-1)/2 \rfloor$ errors. Sphere packing; **perfect** code = spheres fully pack space.
- **Source–channel separation**: optimal error-free comms of a process V over p(Y|X) is achieved by **separate** source code (compress to H) and channel code. Source coding: entropy = optimal compression; channel coding: MI = optimal transmission.

LECTURE 6 · PROBABILISTIC INFERENCE

- Descriptive vs **inferential** stats (parameter estimation, hypothesis testing). **Frequentist**: true θ exists, behaviour over repeated experiments. **Bayesian**: beliefs about θ updated by data.
- **NHST**: H_0 (null, "business as usual") vs H_1; rejection region R: x∈R ⇒ reject H_0. **Simple**: $\theta=\theta_0$ (point); **composite**: $\theta\in\Theta_0$ (region); **nested**: H_0 fixes a parameter of H_1 (a=0 vs a≠0).
- **Type I** (false positive) $\alpha = P(x\in R|H_0)$; **Type II** (false negative) $\beta = P(x\notin R|H_1)$; **power = 1−β**. Mnemonic: boy who cried wolf (I = cried wolf with none). Can only reject / fail to reject H_0 — never "accept".

Neyman–Pearson lemma: for simple H_0:θ_0 vs H_1:θ_1, the most powerful test of size α rejects when $\Lambda(x) = p_{\theta_1}(x)/p_{\theta_0}(x) > k$, with k s.t. $p_{\theta_0}(x\in R)=\alpha$.

- **Proof idea**: any R′ with same α: R\R′ and R′\R have equal H_0-probability φ; on R\R′ $p_{\theta_1} > k p_{\theta_0}$ (mass > φk), on R′\R < $k p_{\theta_0}$ (< φk) ⇒ $p_{\theta_1}(R') < p_{\theta_1}(R)$: R′ has less power.
- **Likelihood ratio (LRT)**

$\Lambda = \sup_{\theta\in\Theta_1} p_\theta(x) \,/\, \sup_{\theta\in\Theta_0} p_\theta(x)$; $\log\Lambda = \log p_{\hat\theta_1}(x) - \log p_{\hat\theta_0}(x)$

optimal for simple H (N-P); for composite w/ monotone likelihood (Karlin–Rubin); otherwise often good, not always optimal. i.i.d.: $\log\Lambda = \Sigma[\log p_{\theta_1^*}(x_i) - \log p_{\theta_0^*}(x_i)]$ → nE[…]. log Λ = "evidence" for H_1 (frequentist). Mock exam 4: plug in MLEs of λ (exp: $\hat\lambda=1/\bar x$) and σ² (half-normal: $\hat\sigma^2=\bar x^2$=mean of squares) into Σ log(q/p).

- **MI as LR test** (H_0: p=f(x)g(y) vs H_1: joint): $\sup_{H_1}$ log-lik = −nH(X,Y); $\sup_{H_0}$ = −n(H(X)+H(Y)) (using $E[\log p_\theta] = -H - D_{KL}$).

$I(X;Y) = (1/n)\, E[\log\Lambda] = E[\log p(x,y)/(p(x)p(y))]$

— MI = evidence for joint over factorised model.

BAYESIAN INFERENCE

$p(\theta|D) = p(D|\theta)p(\theta)/p(D)$ (posterior = likelihood × prior / evidence)
Evidence $p(D) = \int p(D|\theta)p(\theta)d\theta = E_{p(\theta)}[p(D|\theta)]$ ("replace sups with sums")

- Prediction: $p(D'|D) = \int p(D'|\theta)p(\theta|D)d\theta$ (D−θ−D′ Markov chain). **MAP**: θ = argmax $p(D|\theta)p(\theta)$. Posterior = inverse probability problem.
- Gaussian mean (known σ², prior $N(\mu_0,\sigma_0^2)$): $\mu_1 = (\sigma_0^2 x + \sigma^2\mu_0)/(\sigma_0^2+\sigma^2)$, $1/\sigma_1^2 = 1/\sigma_0^2 + 1/\sigma^2$.
- **Bayes factor**

$BF = p(D|H_1)/p(D|H_0)$; posterior odds $p(H_1|D)/p(H_0|D) = BF \times p(H_1)/p(H_0)$

BF≫1 for H_1, ≈1 inconclusive, ≪1 for H_0 (can support either). $p(H|D)=p(D|H)p(H)/p(D)$, $p(D)=\Sigma p(D|H_i)p(H_i)$.

- Coins: H_0:q=½ ⇒ $p(D|H_0)=2^{-n}$; H_1:q~U(0,1) ⇒ $p(D|H_1)=\int q^h(1-q)^{n-h}dq = h!(n-h)!/(n+1)!$; BF = $h!(n-h)!2^n/(n+1)!$. Short n inconclusive; h/n≈½ ⇒ BF→0.
- **Occam's razor**: $p(D|H) \approx p(D|w_{MP},H) \times \sigma_{w|D}/\sigma_w$ = **best-fit likelihood × Occam factor** (low if big prior range but data fits tiny part: flexible models are penalised).
- **MDL** (Shannon's razor): $L(D) = L(H_i) + L(w^*|H_i) + L(D|w^*,H_i)$. Simple H_1: short params, poor fit; complex H_3: opposite; H_2 best tradeoff.

LECTURE 7 · ESTIMATING INFORMATION

- Entropy needs p(x) but we only have samples ⇒ **density estimation**: data → $\hat p(x)$ → plug into information formula.

Method	Idea	Notes
0 Plug-in (discrete)	$\hat p(x_j)$=count/total → $-\Sigma\hat p \log \hat p$	binning continuous data usually bad
I Parametric	fit q_θ (θ^*=argmax $q_\theta(x^n)$), analytic info of fit, e.g. ½ln|2πeΣ|	"garbage in, garbage out": Gaussian fit gives I=0 (Σ=I) for dependent non-Gaussian data; can bound H, I
II Non-parametric	Fit q (e.g. kernel), then $H=-E_p[\log q(x)]$ on the data. KDE: $q(x)=(1/n)\Sigma K_h(x-x_i)$, $K_h=(2\pi h^2)^{-½}e^{-(x-x_i)^2/2h^2}$	pick h by cross-validation; k-NN (Kraskov): adaptive width, area A with K neighbours, locally uniform — more accurate
III Variational	estimation → optimisation: $H=\max_\theta \hat H_\theta(x^n)$	scalable, deep learning

- **Variational MI (predictor, Barber–Agakov)**:

$I(X;Y) \ge H(Y) + (1/n)\Sigma \log q_\theta(y_i|x_i)$

(Gibbs; any q_θ, e.g. $N(y;f_\theta(x),\sigma^2)$); good when Y low-dim/discrete.

- **Contrastive (density ratio)**: positives $(x_i,y_i)\sim p(x,y)$, negatives from p(x)p(y) (random pairing); train classifier f_θ; $I \ge L_{CE}$; optimal f = p(x,y)/(p(x)p(y)).
- **Bias** = $\langle\hat I\rangle - I$; **Variance** = $\langle(\hat I - \langle\hat I\rangle)^2\rangle$; both worsen with limited data; estimators trade them off. Cheaper methods make more assumptions.
- **Hypothesis testing in practice**: statistic T(x); **sufficient** if θ−T−X Markov chain (e.g. Λ, $\hat I$). R={x: T(x)>k}. Q1 significance testing; Q2 power analysis.
- **p-value** = P(T ≥ observed | H_0); significant at level α if p<α (usual 0.05). Null distribution $p_{\theta_0}(T=t)$; critical value $\alpha=\int_{t_crit}^{\infty} p_{\theta_0}(t)dt$. Report more than just p-values! **Wilks**: 2logΛ → χ² as n→∞ (finite n hard analytically).
- **Surrogate data**: sample the null empirically by generating fake data where H_0 holds, removing only the property tested. Independence test: ① compute $\hat I(X;Y)$; ② shuffle $Y\to Y^S$; ③ $\hat I(X;Y^S)$ for many shuffles; ④ $p = \#\{\hat I(X;Y^S) \ge \hat I(X;Y)\} / \#\text{surrogates}$. Triplet: statistic (MI) + null (X⊥Y) + surrogate (shuffle).
- **Power analysis**: sample size needed for effect; power=1−β=P(x∈R|H_1) (true-positive rate). Simulate: pick n & μ>0 under H_1, draw n samples, run NHST at α, repeat M times ⇒ power = fraction rejected. Power ↑ with n and effect size. Analytic via CDF/inverse CDF; info quantities ~ non-central χ²(λ). **Bias estimation**: fit p(x,y) to data → ground-truth I → draw M datasets of size n → $\hat I$ → compare.

LECTURE 8 · INFORMATION GEOMETRY

- **Manifold**: locally like $\mathbb{R}^n$; small steps $ds^2=dx^T g(x) dx$ with metric g (polar: g=diag(1,r²)). Euclid: $s^2=z^TGz$, G=I. **Statistical manifold**: points = PDFs; same manifold, different coordinates (variance vs precision $\Lambda=\Sigma^{-1}$).
- **Divergence** D: ① D(p,q)≥0; ② D=0 iff p=q; ③ D(p,p+dp) positive-definite quadratic form. KL satisfies all (others: Bregman, f-divergences).
- **Projection**: idempotent P=P∘P (Euclid: Proj v = argmin‖v−w‖²).

I-projection: $q^*_I = \text{argmin}_{q\in Q} D_{KL}(q\|p)$ (zero-forcing; E wrt q — approximate complex p by simple q)
M-projection: $q^*_M = \text{argmin}_{q\in Q} D_{KL}(p\|q)$ (zero-avoiding; E wrt p — real data)

- **MLE = M-projection**: $D(p\|q) = -H(p) - E_p[\log q] \approx -H(p) - (1/n)\Sigma \log q(x_i)$.
- **MI = M-projection** onto factorisable F={q: q(x|y)=q(x)}:

$I(X;Y) = \min_{q\in F} D_{KL}(p\|q)$

(min at f=p(x), g=p(y) by Gibbs ⇒ −H(X,Y)+H(X)+H(Y)). Cond. MI: M-proj onto C={q(x|y,z)=q(x|z)}: I(X;Y|Z).

- Pythagoras: r = M-proj of p on Q, $q\in Q \Rightarrow D(p\|q) = D(p\|r) + D(r\|q)$

(also I-proj; needs "dually flat" Q). Applied to F⊂C ⇒ I(X;Y,Z)=I(X;Y|Z)+I(X;Z) (= **MI chain rule**).

- **Max-entropy principle** (Jaynes): given $E[f_i(X)]=\alpha_i$, choose max H subject to them. Lagrange: $\mathcal{L}=-\Sigma p \log p + \Sigma\lambda_i(\Sigma p\, f_i - \alpha_i)$ ⇒

$p(x) = (1/Z)\exp(-\Sigma_i \lambda_i f_i(x))$ (exponential family)

λ set s.t. constraints hold. Maxent = **I-projection of uniform u** on Q={$E_q[f_i]=\alpha_i$} ($D(q\|u)=-H(q)+\log u$). More constraints ⇒ $H(q_2^*) \le H(q_1^*)$.

- **Fisher information**: $D(\theta\|\theta+d\theta) \approx ½ d\theta^T I(\theta)\, d\theta$ (offset & 1st derivative vanish by Gibbs).

$I(\theta) = -E[\partial^2 \log p_\theta(X)/\partial\theta^2] = \text{Var(score)} = E[(\partial \log p_\theta/\partial\theta)^2]$, score = $\partial \log p_\theta/\partial\theta$

Vector: $I_{jk} = -E[\partial^2 \log p/\partial\theta_j \partial\theta_k]$ (Hessian) — natural metric; high I = sharp likelihood peak, KL grows fast. Gaussian (μ,σ): $I = \text{diag}(\sigma^{-2}, 2\sigma^{-2})$.

- **Bias–variance**: $MSE(\hat\theta)=E[(\theta-\hat\theta)^2]$=**Bias² + Var** (cross term vanishes since $E[\hat\theta - E\hat\theta]=0$).

Cramér–Rao: $MSE(\hat\theta) \ge b(\theta)^2 + (1+b'(\theta))^2/I(\theta)$

(b = bias). More curved p_θ ⇒ easier estimation.

- **Wrap-up**: same formulae derived 3 ways — optimal communication, optimal hypothesis tests, projections on manifolds. Info theory sets limits of what is possible in ML.